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SUMMARY

Second-year Computer Science student at NUCES with practical experience in backend development, Linux-based automation, and container technologies. Proficient in Python, C++, SQL, and Docker, with a growing interest in DevSecOps and intelligent systems. Enjoys developing efficient and secure software solutions, and is currently looking for an internship opportunity to further enhance skills through real-world projects.

EXPERIENCE

- Gained hands-on experience in networking using Bash Script, Linux, and Wireshark.
- Used tools like Docker, Jenkins, Git & GitHub for day-to-day DevOps workflows.

PROJECTS

Smart City Traffic & Emergency Response AI System — Python, Matplotlib

- Built a full-featured AI simulation integrating ANN priority prediction, a Logic Knowledge Base, a CSP traffic signal scheduler, and BFS/UCS/A* route-finding into a single unified pipeline with a 6-tab matplotlib GUI.
- Designed modular architecture across 7 components (preprocessor, router, ANN, KB, CSP, navigation, response) supporting 5 request categories including Emergency_Response and Integrated_City_Service workflows.

Smart City Transport System — Oracle 11g, Python/Flask, HTML/CSS/JavaScript

- Built a full-stack transport management system for the Faisalabad region with Oracle 11g as the core database handling passenger ticketing, route management, vehicle scheduling, and administrative reporting.
- Implemented advanced SQL/PL/SQL features including complex joins, subqueries, views (e.g., ActiveRoutes_VW), automated triggers, and stored procedures to ensure strict data integrity.
- Developed a Python/Flask backend API connected to a custom HTML/CSS/JavaScript frontend for seamless end-to-end data flow.

GridPath-AI — Python, Tkinter

- Built a real-time grid-based pathfinding visualizer implementing BFS, DFS, and Bidirectional Search algorithms.
- Designed an interactive Tkinter GUI allowing users to place walls and visualize traversal step-by-step.

CDAF (Discrete Algorithms Framework) — C++

- Implemented custom data structures including sets, dynamic arrays, and relations entirely without STL.
- Built graph algorithms from scratch including Dijkstra and Kruskal for minimum spanning trees.

DSA Visualizer — C++, WebAssembly (Emscripten)

- Developed a high-performance web-based visualizer covering AVL Trees, Heaps, Graphs, and hashing mechanisms.
- Compiled C++ to WebAssembly using Emscripten, enabling browser-native execution without plugins.

Atari Breakout Game — x86 Assembly

- Wrote a fully functional arcade game in x86 Assembly with memory-level control over rendering and input.
- Implemented a power-up system and dynamic difficulty scaling for a complete gameplay experience.

File Organizer Automation Script — Bash, Linux

- Built an automated file sorting system that organizes files by extension using CLI argument handling.
- Designed for easy extensibility and scheduled execution via cron for hands-free maintenance.

SKILLS

Languages: Python, C/C++, SQL, JavaScript, HTML/CSS, Bash

Frameworks & Libraries: React, Tkinter, NumPy, Matplotlib

DevOps & Tools: Docker, Jenkins, Git, GitHub, Linux, AWS

Developer Tools: VS Code, Visual Studio, Jupyter Notebook, Oracle

EDUCATION

BS Computer Science September 2024 – June 2028

National University of Computer and Emerging Sciences | CGPA: 2.92 / 4.0